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EXERCISE 4: FREQUENCY SELECTIVITY AND MASKING

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1. INTRODUCTION

The auditory system responds selectively to the spectral content of the stimulus. The individual spectral components of the stimulus can be perceived distinct if their spectral separation is sufficiently large. This property of the auditory system is known as "frequency selectivity" and is assumed to be due to the tonotopical organization of the basilar membrane. When different sound sources are present (e.g., a pure tone and background noise), they will mask each other if their spectral separation is finer than the frequency selectivity of the ear. More precisely, masking will occur if the spectral separation is so small that spectral components of different sources fall within the same auditory filter. In the case of a tone-in-noise detection task, if the frequency selectivity of the ear is not sufficient for separating the tone from the background noise, masking will occur and a higher level of the tone will be necessary for detection (i.e., a higher detection threshold will be measured). A first estimate of the auditory filter bandwidth (and therefore of frequency selectivity) is given by the so-called "critical band" (CB).

In this exercise we will perform different types of masking experiments and learn how to estimate the auditory filter bandwidth with different methods. A strong assumption in this exercise is that the auditory system is considered linear.

1.1. Exercise overview

In the first part of the exercise, the power-spectrum model by [1] will be explained and you will conduct the so-called band-widening experiment to measure the masked threshold of a pure tone in bandpass noise with different bandwidths. In this part of the exercise, the shape of the auditory filter is assumed to be rectangular. From the experimental results, you will estimate the width of the CB and calculate the listener-specific detection efficiency k. Then, you will derive a formula for estimating a generic (non rectangular) shape of the auditory filter from the measured masked thresholds.

In the second part of the exercise, you will measure the auditory filter shape and bandwidth with the so-called notched-noise experiment. This method was introduced to overcome some of the drawbacks associated with Fletcher's original approach. Here the simple assumption of a rectangular filter is replaced by a more realistic filter model, namely the "rounded-exponential filter" (roex). You will estimate the shape of your own auditory filter by fitting the parameters of the roex filter to the data measured in the experiment. The fitted auditory filter can be used to predict the data measured during Fletcher's experiment.

Finally, the concept of the equivalent rectangular bandwidth (ERB) is introduced and compared with the critical band estimate obtained from Fletcher's experiment.

1.2. Exercise objectives

After completion of this lab exercise and the associated report you will be able to:

- Describe the power-spectrum model of masking with its assumptions and relate it to the critical-band concept.
- Calculate the masking threshold predicted by the power-spectrum model for different noise and auditory-filter configurations.
- Explain the relation between the masking thresholds and the shape of the auditory filter.
- Explain the difference between Fletcher's experiment and the notched-noise method regarding the estimation of an auditory-filter shape.
- Estimate your individual auditory filter shape and bandwidth, calculate its ERB, and evaluate it in comparison with other measurements, both your own and from literature.

2. THE POWER-SPECTRUM MODEL

2.1. Theoretical Background

[1] conceptually modelled frequency selectivity in the auditory system as a set of bandpass filters (the auditory filters) that separate the spectral information of the stimulus into different channels. According to his model, different frequency components mask each other only if they fall within the same auditory filter. The band-widening experiment, allows us to observe this phenomenon and obtain a first approximation of the auditory-filter bandwidth. The experiment consists of measuring the detection threshold of a pure tone (the signal) in the presence of a band-limited noise masker as a function of noise bandwidth. The noise is spectrally centred on the

signal frequency and its spectral density is constant across frequency (cf. Fig. 1), thus the overall power of the noise increases with increasing noise bandwidth. Fletcher observed that the detection threshold of the pure tone increased only for small noise bandwidths and remained approximately constant when the bandwidth exceeded a certain value, the CB (for an overview, see [2]). It is hypothesised that only the amount of spectral energy that falls within the auditory filter centred at the pure-tone frequency contributes to masking. In the experiment, the noise energy falling within the auditory filter increases for increasing noise bandwidth, until the auditory filter is saturated. The amount of noise energy inside the auditory filter remains constant for any bandwidth broader than the CB and so does the detection threshold.

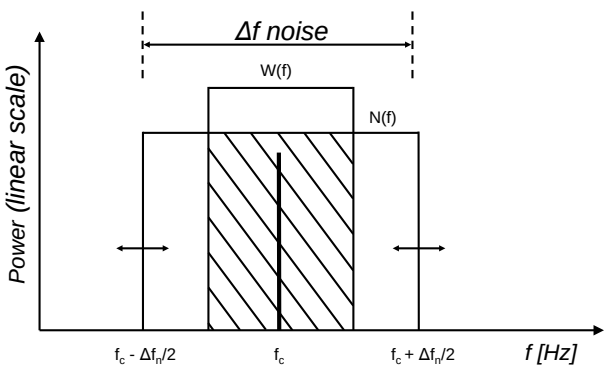

Fig. 1. Schematic illustration of Fletcher's band-widening experiment. The detection threshold of a pure tone signal is measured as a function of the noise bandwidth Δf . The amount of noise at the output of the critical band centred at the signal frequency is indicated by the shaded area.

The phenomenon observed in the band-widening experiment can be explained by the power-spectrum model of masking. In this model, the signal and the masker are represented by their long-term power spectra, P_s and $N(f)$, respectively. The model assumes a constant ratio between signal and noise powers at the output of the auditory filters:

$$k = \frac{P_s^{\text{post}}}{N^{\text{post}}}\,, \tag{1}$$

where P_s^{post} and N^{post} denote the signal and noise power after filtering. The detection threshold corresponds to a signal-to-noise ratio k that may be regarded as a measure of the efficiency of the detection process. The efficiency k can be influenced by threshold criterion, practice or inattention, and is assumed to be fixed for a given experimental condition. For Fletcher's experiment, k is typically in the range of 0.25 to 0.5. Smaller k values indicate that the listener is more sensitive in the detection task.

According to Eq. (1), the power of the signal at threshold is directly proportional to the noise power that falls into the auditory filter centred at the signal frequency f_c. Introducing the power transfer function $W(f)$ of the auditory filter (squared modulus of the filter transfer function), the masking equation becomes

$$P_s = k \int_{-\infty}^{\infty} W(f) N(f) \mathrm{d}f\,, \tag{2}$$

where we assumed $P_s^{\text{post}} = P_s$, i.e., zero filter attenuation at the signal frequency. For the Fletcher experiment, the power spectral density $N(f)$ is constant at N_0 within the passband of bandwidth Δf. Then, assuming the auditory filters to be of rectangular shape with bandwidth Δf_{CB}, Eq. (2) simplifies to

$$P_s(\Delta f) = k\, N_0 \cdot \min\{\Delta f_{CB}, \Delta f\}\,. \tag{3}$$

2.2. Experiment

A band-widening experiment according to Fletcher is provided within the MATLAB AFC framework and is called "fletcher". Please refer to Appendix A for a description of how to start an experiment and how to plot and access the data. The experiment is set up with a **target tone at 2000 Hz** and is measured for **6 noise bandwidths** ranging from **50 to 1600 Hz** in octave steps. The **power spectral density** of the noise is **40 dB SPL/Hz**. The stimuli are presented to the left ear only. An adaptive, three-interval, three-alternative forced-choice method combined with a one-up two-down tracking rule is used. This procedure is used in all experiments of this exercise. At least two listeners should obtain a set of data (6 points, time: about 9 min. per person). In order to collect more than one data point you are required to continue with the next measurement when the procedure asks whether you want to start the next measurement. The procedure will indicate when the experiment is finished.

2.3. Questions

- **[BACKGROUND]** Can you derive Eq. (3)?
- **[BACKGROUND]** How is the overall presentation level of the noise for a given bandwidth calculated from the power spectral density? Calculate the values in dB SPL for the bandwidths used in the experiment. Note: Keep in mind that P_s and N_0 in Eqn. (3) are not expressed in units of dB.
- **[RESULTS]** What do you expect the threshold curve $P_s(\Delta f)$ for the Fletcher experiment to look like, assuming that $W(f)$ is a rectangular function?
- **[RESULTS]** Estimate the critical bandwidth Δf_{CB} from your data.
- **[RESULTS]** Estimate k from the data points above the critical bandwidth Δf_{CB}.
- **[BACKGROUND]** Find an expression based on Eqn. (2) that describes $P_s(\Delta f)$ for the Fletcher experiment

without any presumptions about the filter shape, i.e., for a general filter power transfer function $W(f)$. (Hint: Consider where the noise spectral density is non-zero.) Next, simplify the resulting equation by assuming the filter to be symmetric around (and centred at) the target-tone frequency f_c. Solve for W using the Leibniz integral rule given in section B.1 (Final result: $W(f_c \pm \frac{\Delta f}{2}) = \frac{1}{kN_0} \cdot \frac{\mathrm{d}P_s}{\mathrm{d}\Delta f}$).

- **[DISCUSS]** What do you think about the accuracy of this method to derive the filter shape?

3. ESTIMATING THE SHAPE OF THE AUDITORY FILTER

The band-widening experiment gave us an estimate of the bandwidth of the auditory filter. Like Fletcher's experiment, most methods for estimating the auditory-filter shape are based on the power-spectrum model. A major problem in masking experiments that aim at estimating the auditory-filter shape is that the auditory system might benefit from using a filter that is not centred at the signal frequency. This happens when the experimental design allows the auditory system to gain a better signal-to-noise ratio by slightly shifting the filter. This phenomenon is called "off-frequency" listening and makes it difficult to estimate the shape of a single auditory filter. Off-frequency listening is a potential issue in Fletcher's band-widening experiment (assuming a filter with steep slopes and a rather flat tip, i.e., a rectangular filter). Another problem with Fletcher's experimental design relates to estimating the thresholds around the filter tip. In this case, as you might have experienced in the experiment, very narrow-band noise maskers are used which typically show strong fluctuations due to beating. This corrupts the assumption that the noise is stationary and may therefore bias the detection thresholds.

3.1. The notched-noise method

A method for determining the auditory filter shape which prevents off-frequency listening as well as beating effects is the notched-noise method[1]. An illustration of the spectral configuration of the signal is given in Fig. 2. The signal (bold vertical line) is fixed in frequency and the bandwidth of the spectral notch ($2\Delta f$) is varied. Note that we now use $2\Delta f$ for the bandwidth of the notch instead of Δf as in the Fletcher experiment in order to simplify the mathematical treatment of the problem. The deviation of each edge from the signal frequency is Δf. Since the notch is symmetric around the signal frequency, the method assumes symmetric auditory filters. This assumption appears reasonable at medium levels.

[1]For additional background information and guidance with some of the next questions, see [3], which can be found in the reader.

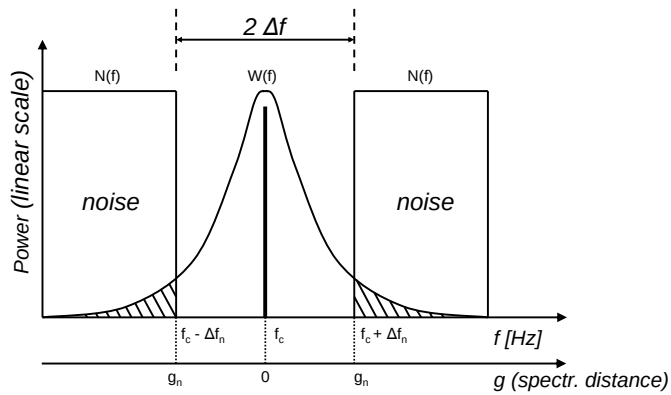

Fig. 2. Schematic illustration of the notched-noise method. The threshold of the sinusoidal signal is measured as a function of the width of the spectral notch in the masker. The amount of noise at the output of the auditory filter centred at the signal frequency is indicated by the shaded areas. The lower abscissa shows the spectral distance measure g, see (Eqn. (7)). The g-axis is obtained by shifting and scaling the frequency- axis. Note that g is unitless!

For this notched-noise configuration, the maximum signal-to-noise ratio is always achieved in the filter centred at the signal frequency. In terms of the power-spectrum model, the threshold is proportional to the shaded area under the filter curve in Fig. 2. The notched noise with spectral density N_0 modifies the general power-spectrum model of masking Eqn. (2) to

$$P_s(\Delta f) = kN_0 \left(\int_0^{f_c - \Delta f} W(f)\mathrm{d}f + \int_{f_c+\Delta f}^{f_u} W(f)\mathrm{d}f \right) , \tag{4}$$

where f_u denotes the upper cut-off frequency of the upper band. Due to symmetry of the filter function and the noise spectrum, we can simplify Eqn. (4) to:

$$P_s(\Delta f) = 2kN_0 \int_{f_c+\Delta f}^{f_u} W(f)\mathrm{d}f , \tag{5}$$

With the assumption that $W(f)$ is already close to zero at f_u, this can be simplified to

$$P_s(\Delta f) = 2kN_0 \int_{f_c+\Delta f}^{\infty} W(f)\mathrm{d}f . \tag{6}$$

Since $W(f)$ is assumed to be symmetric around the centre frequency, it is convenient to measure frequency in terms of the relative spectral distance from the centre. The new frequency variable, g, is given by

$$g = \frac{\Delta f}{f_c} = \frac{|f - f_c|}{f_c} . \tag{7}$$

With this, Eq. (6) can be reformulated as

$$P_s(g_n) = 2kN_0 f_c \int_{g_n}^{\infty} W(gf_c + f_c)\mathrm{d}g , \tag{8}$$

where $g_n = (|f_n - f_c|)/f_c$ and the subscript n refers to an arbitrary notched-noise condition. In the following, the filter function $W(gf_c + f_c)$ is denoted as $\widehat{W}(g)$, such that Eqn. (8) modifies to

$$P_s(g_n) = 2kN_0 f_c \int_{g_n}^{\infty} \widehat{W}(g)\mathrm{d}g \ . \tag{9}$$

3.2. Experiment

The notched-noise experiment is provided within the MATLAB AFC framework and is named "notchedNoise". The experiment is set up with a **target tone at 2000 Hz**. **Six** different **notched-noise conditions** are measured, with $g_n = 0, 0.05, 0.1, 0.2, 0.3, 0.4$. Like in Fletcher's experiment, the **spectral density** of the noise is **40 dB SPL/Hz**. The same adaptive procedure is used. Again, at least two listeners should obtain a set of data (6 points, time: about 10 min per person). In order to be able to compare the results to the first experiment, at least one person should participate in both measurements.

3.3. The rounded-exponential filter

As you will see, the data from the notched-noise experiment follow, to a good first approximation, a straight line when plotted in dB (logarithmic scale) against $\Delta f/f_c$ (linear scale). A deviation from the straight line is observed for small notch widths, where the threshold curve is flatter. [4] have suggested an exponential filter shape with a "rounded" tip, the "rounded-exponential" filter. The simplest filter form (only one parameter) is given by

$$\widehat{W}(g) = (1 + pg)e^{-pg} \ , \quad g \geq 0 \ , \tag{10}$$

where g is the normalized distance to the centre frequency f_c of the filter $g = (|f - f_c|)/f_c$. The filter is symmetrical around f_c and looks like a pair of back-to-back exponentials with a rounded tip. The parameter p determines the width of the passband and the slope of the filter skirts. Assuming the basic rounded-exponential filter shape given in Eqn. (10), Eqn. (9) obtained for the notched-noise experiment can now be solved as

$$P_{s,notched}(g_n) = 2kN_0 f_c \left(g_n + \frac{2}{p}\right) e^{-pg_n} \ . \tag{11}$$

Similarly, by performing an appropriate change of the integration limits, the threshold curve for Fletcher's experiment calculates to

$$P_{s,Fletcher}(g_n) = 2kN_0 f_c \left(\frac{2}{p} - e^{-pg_n}\frac{2 + pg_n}{p}\right) \ . \tag{12}$$

You will need these functions to fit the parameter p of the roex filter and the parameter k of the power spectrum model to your data to obtain your own auditory filter.

In order to compare the bandwidth of the resulting roex filter to the estimate of the CB from Fletcher's experiment, one needs to quantify the bandwidth of the roex filter. A common way to characterize the bandwidth of a non-rectangular filter in psychoacoustics is the equivalent rectangular bandwidth (ERB) of the filter. This is the bandwidth of a perfectly rectangular filter that passes the same power as the filter under test when excited with all frequencies. [5] have fitted human data on the equivalent rectangular bandwidth (ERB) of the auditory filter at a centre frequency f_c (in Hz) with the function

$$\mathrm{ERB} = 24.7 + 0.108\, f_c \ . \tag{13}$$

3.4. Questions

- **[BACKGROUND]** Can you derive Eq. (8)? You might want to refer section B.2 on substitution of an integration variable.
- **[BACKGROUND]** Can you derive Eq. (11)[2] and Eq. (12)?
- **[RESULTS]** In the notched-noise experiment the data follows (approximately) a straight line when plotted in dB against $\Delta f/f_c$. Why does this imply that the signal threshold is an exponential function of the notch width, and so is the auditory filter shape. (Keep in mind the relation between threshold curve and filter shape in Eq. (9).)
- **[IMPLEMENT]** Implement Eqn. (10) as a MATLAB function `RoexFilter(g,p)` that outputs a vector `w` for any input vector `g` and the parameter `p`. Edit the existing function `RoexFilter.m` in `FSaM\scripts`.
- Implement Eqn. (11) as a MATLAB function `RoexFilterTailPSM(gn,N0dB,fc,p,k)`, where `N0dB` is the spectral density in dB. `RoexFilterTailPSM` has to return a threshold value in dB, which is $10\log[P_{s,notched}(g_n)]$. Edit the existing function `RoexFilterTailPSM.m` in `FSaM\scripts`.
- **[IMPLEMENT]** Use the command `pldata` with the optional output parameters to read in your data and the command `RoexFilterTailPSMFit` to perform the fitting (see Sec. A). You should plot the threshold prediction using the fitted values of p and k, and compare it to your data. You should also plot the resulting roex filter.
- Use the values for p and k you calculated from the notched-noise data to predict the data obtained earlier in the Fletcher experiment. To do so, replace Eq. (11) -

[2] The following indefinite integral might be of use: $\int xe^{\alpha x}\,\mathrm{d}x = \frac{1}{\alpha^2}(\alpha x - 1)\,e^{\alpha x}$.

which yields $P_{s,notched}(g_n)$ - in `RoexFilterTailPSM(gn,N0dB,fc,p,k)` by Eq. (12), such that it yields $P_{s,Fletcher}(g_n)$. Compare the predicted thresholds to your empirical data.

- **[DISCUSS]** Compare your fitted value of k to your estimate from Fletcher's experiment.
- **[BACKGROUND]** Find an expression for the ERB of the roex filter. Hint: Eq. (11) for $P_{s,notched}(g_n)$.
- **[DISCUSS]** Is your result for the ERB at $f_c = 2000$ Hz consistent with the expression in Eq. (13)? If not, what could be possible reasons for the discrepancy?
- **[DISCUSS]** Would you expect the ERB estimate to change if the noise spectral density in the notched-noise experiment was raised or lowered? Motivate your answer.

4. REFERENCES

[1] H. Fletcher, "Auditory patterns," *Reviews of Modern Physics*, vol. 12, pp. 47–65, 1940.

[2] B. C. J. Moore, *An Introduction to the Psychology of Hearing*, chapter Frequency selectivity and masking, Academic Press, 1997.

[3] R. D. Patterson and B. C. J. Moore, *Frequency Selectivity in Hearing*, chapter Auditory filters and excitation patterns as representations of frequency resolution, Academic, London, 1986.

[4] R. D. Patterson, I. Nimmo-Smith, D. L. Weber, and R. Milroy, "The deterioration of hearing with age: frequency selectivity, the critical ratio, the audiogram, and speech threshold," *J. Acoust. Soc. Am.*, vol. 72, pp. 1788–1803, 1982.

[5] B. R. Glasberg and B. C. J. Moore, "Derivation of auditory filter shapes from notched-noise data," *Hear. Res.*, vol. 47, pp. 103–138, 1990.

A. AFC MATLAB TOOLS

A.1. Starting measurements

To start a measurement you have to use the `afc_main` command:

```
afc_main(expname, ts, group);
```

The input parameters are as follows:

- `expname:` The name of the experiment
- `ts:` The initials of the test subject
- `group:` Your group number (must be in the form 'groupN')

For example, to start the experiment 'fletcher' for subject 'Anon Ymous', member of group 3, enter the following command:

```
afc_main('fletcher','ay','group3');            % This is just an example!
```

Use the number block to answer the questions during the measurement. Make sure that you have activated 'NumLock' (check the NumLock-LED). If the experiment consists of more than a single experimental condition (e.g., different noise bandwidths) you are asked to continue with the next measurement until the procedure indicates that the experiment is finished.

A.2. Plotting the results

The data of all the measurements will be saved in your current working directory[3]. The different filenames follow the scheme

```
expname_ts_group.dat
```

Thus, the data for the example above could be found in the file `fletcher_ay_group3.dat`. To plot the results of AFC measurements, use the command `pldata`:

```
[x,y,ystd] = pldata(expname, ts, group [, proc, allmean]);
```

The input parameters are almost the same as described above:

- `expname:` The name of the experiment
- `ts:` The initials of the subject who did the measurement. To see results of more than one subject, enclose the initials with curly brackets `{}`. Examples:
  - `'ay'`
  - `{'ay','bm','co'}`
- `group:` The name of your group
- `proc:` Optional Parameter. Possible values:
  - `'mean'` Plot the mean data (within subject)
  - `'raw'` Plot the raw data (default)
- `allmean:` Optional Parameter. Possible values:
  - `'yes'` Plot the overall mean across subjects.
  - `'no'` Only plot individual results (default)

The (optional) output parameters are:

- `x:` The experimental parameter; in our case band or notch width
- `y:` The results of the experiment; in our case level at threshold in dB
- `ystd:` The standard deviation of `y`

For example, to plot and return the results from subject 'ay' for the experiment above, you can use:

```
[x,y,ystd] = pldata('fletcher', 'ay', 'group3');    % Again, only an example.
```

Note: For the Fletcher experiment the noise bandwidth (abscissa) is plotted on a logarithmic scale. If you want to plot your own threshold prediction in the same figure, you should transform your bandwidth (given in Hz) by plotting the log10 of the bandwidth instead.
E.g.: `plot(log10(bandwidth_Hz),Ps)`, where `Ps` is your threshold prediction.

[3] If you changed the directory (e.g., by using the `cd` command) you can get the actual working directory with the `pwd` command.

A.3. Fitting the roex filter to notched-noise data

After you wrote a MATLAB script `RoexFilter.m` and `RoexFilterTailPSM.m`, use the command `RoexFilterTailPSMFit` to fit the roex filter to the data of your notched-noise measurements:

```
RoexFilterTailPSMFit.m - Fits single parameter roex(p) to
experimental notched-noise data.

Usage: [p,k] = RoexFilterTailPSMFit(xdata,thres,N0dB,fc)

xdata  = column vector of deltaf/fc used in experiment
thres  = column vector of threshold values at xdata
N0dB   = spectral density of noise
fc     = center frequency of filter

p      = fitted slope parameter p of roex(p)
k      = fitted power spectrum model proportionality constant k

See also: RoexFilter, RoexFilterTailPSM
```

Use the outputs of `pldata.m` to access your data. If a figure generated by `pldata` is still existing you can add a new trace for, e.g., your threshold predictions with the roex filter, by using the MATLAB `plot` command. `pldata` sets the figure to "hold" (see `help hold`).

B. MATHEMATICAL ANNOTATIONS

B.1. Leibniz integral rule

The Leibniz integral rule gives a formula for the differentiation of a definite integral:

$$\frac{\partial}{\partial z}\left(\int_{a(z)}^{b(z)} W(f)\,\mathrm{d}f\right) = W(b(z))\cdot\frac{\partial b(z)}{\partial z} - W(a(z))\cdot\frac{\partial a(z)}{\partial z}\,. \tag{14}$$

B.2. Substitution of an integration variable

Given an integral $\int_a^b W(f)\,\mathrm{d}f$, one can substitute the integration variable f by another as follows. Assume the new variable g is given as a function of f: $g = G(f)$. Then $\frac{\mathrm{d}g}{\mathrm{d}f} = G'(f)$, hence:

$$\mathrm{d}f = \frac{1}{G'(f)}\mathrm{d}g \tag{15}$$

With the help of the inverse function G^{-1} we may express f as a function of g:

$$f = G^{-1}(g) \tag{16}$$

Thus, we get: $\mathrm{d}f = \frac{1}{G'(G^{-1}(g))}\,\mathrm{d}g$. Substituting this in the original integral we arrive at the result:

$$\int_a^b W(f)\,\mathrm{d}f = \int_{G(a)}^{G(b)} W(G^{-1}(g))\frac{1}{G'(G^{-1}(g))}\,\mathrm{d}g\,, \tag{17}$$

where f has been replaced by the new variable g. In the present context $W(G^{-1}(g))$ is just the power transfer function $\widehat{W}(g)$.